

Multimodal Deep Fusion for Antarctic Sea Ice Classification

Project Summary Report

Research Seminar -- Knox College

Best Model (Deep Fusion): mIoU = 0.9010 | Macro F1 = 0.9468 | Pixel Acc = 95.3%

This report summarizes a multimodal deep-learning framework that performs per-pixel classification of Antarctic sea ice. The framework fuses Sentinel-2 optical imagery with ICESat-2 ATL03 photon-altimetry data through a deep feature-level fusion stage combining a U-Net image branch (ResNet-18 encoder) and a recurrent LSTM photon branch. On a geographically held-out test tile (T03CWT), the fused model significantly outperforms both unimodal baselines, especially on minority classes (thin ice and open water).

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1. Data Sources and Problem Setup

The sea ice classification task is formulated as per-pixel prediction of three surface classes: thick ice, thin ice, and open water. Models are trained on tiles T02CNA and T02CNC, and evaluated on the geographically separated tile T03CWT, providing a strict test of geographic generalization.

Source	Description	Resolution
Sentinel-2 L1C	Optical RGB imagery, Ross Sea region	10 m
ICESat-2 ATL03	Photon altimetry, aggregated to along-track segments	10 m along-track
Ground-truth	HSV-thresholded per-pixel labels (3 classes)	10 m

Class encoding in masks: red = thick ice, blue = thin ice, green = open water. Eight photon features are extracted per segment: `h_cor_mean`, `h_cor_med`, `h_diff`, `rel_height_min_elev`, `height_sd`, `pcnth_mean`, `pcnt_mean`, `bcnt_mean/brate_mean`.

2. Model Architecture

2.1 Image Branch (U-Net)

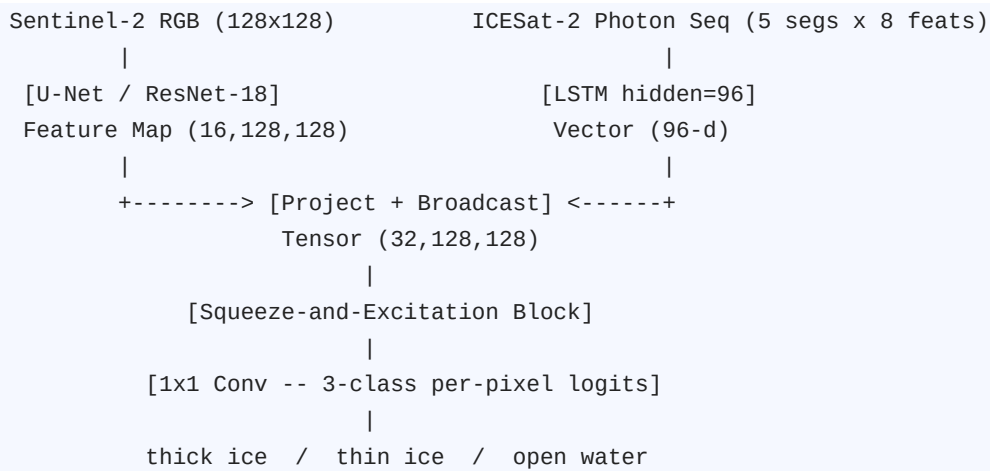
Each 128 x 128 RGB patch is processed by a U-Net with a ResNet-18 encoder pretrained on ImageNet. The decoder restores spatial resolution and produces a 16-channel feature map (16, 128, 128).

2.2 Photon Branch (LSTM)

A sliding window of 5 consecutive ATL03 segments forms the input sequence. A single-layer LSTM (hidden dim = 96, dropout = 0.4) followed by fully connected layers and a softmax head predicts the surface class at each labeled point. Training uses categorical focal loss: $\alpha = [0.05, 0.45, 0.60]$, $\gamma = 2.0$.

2.3 Fusion Stage

The 96-d photon vector is projected to 16 channels and broadcast spatially, then concatenated with the U-Net feature map to form a 32-channel tensor. A Squeeze-and-Excitation block (reduction ratio = 8) recalibrates channels. A 1 x 1 convolution produces three-class per-pixel logits. The photon branch is fine-tuned at 0.1x the base learning rate during fusion.



3. LSTM Hyperparameter Sweep

A 21-configuration sweep selected the best photon-branch settings before fusion. Parameters swept: focal loss alpha weights, focal loss gamma (1.0/2.0/3.0/5.0), hidden dimension (32/64/96), learning rate (5e-4 to 2e-3), dropout (0.2-0.5), sequence length (3/5/7), and random seed (7/42/123).

Winning configuration: hidden = 96, dropout = 0.4, alpha = [0.05, 0.45, 0.60], gamma = 2.0, seq_len = 5, lr = 8.9e-4, seed = 42. This yielded LSTM mIoU = 0.6978 on the held-out tile T03CWT.

4. Results

4.1 Per-Model Summary (test tile T03CWT)

Model	Input Modality	mIoU	Pix. Acc.	Macro F1
U-Net	Sentinel-2 optical	0.8704	0.9429	N/A
LSTM	ICESat-2 photon	0.6978	0.9594	0.8080
Deep Fusion	optical + photon	0.9010	0.9530	0.9468

4.2 Per-Class IoU

Model	Thick Ice	Thin Ice	Open Water
U-Net	0.9299	0.7683	0.9130
LSTM	0.9671	0.5427	0.5836
Deep Fusion	0.9403	0.8138	0.9489

The fusion model yields the largest improvements on minority classes: thin-ice IoU increases from 0.768 to 0.814 (+4.5 pp over U-Net); open-water IoU increases from 0.913 to 0.949 (+3.6 pp over U-Net).

4.3 Deep Fusion Per-Class Precision, Recall, F1

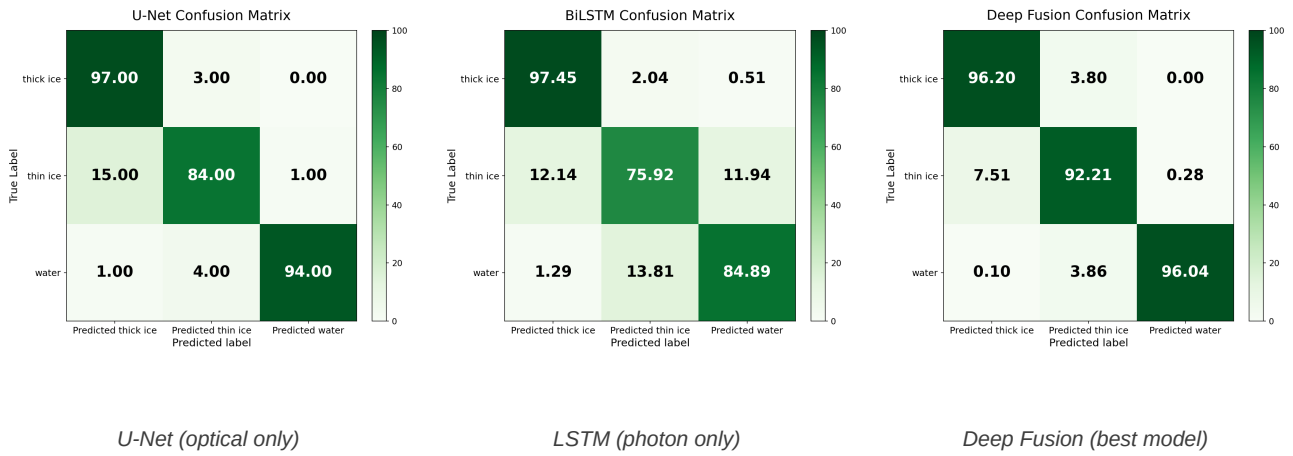
Class	Precision	Recall	F1
Thick Ice	0.9766	0.9620	0.9692
Thin Ice	0.8739	0.9221	0.8973
Open Water	0.9876	0.9604	0.9738

4.4 LSTM Per-Class Precision, Recall, F1

Class	Precision	Recall	F1
Thick Ice	0.9922	0.9745	0.9833
Thin Ice	0.6555	0.7592	0.7036
Open Water	0.6512	0.8489	0.7370

5. Confusion Matrices

Row-normalized confusion matrices (percentage of true-class pixels classified per predicted class) on the held-out test tile T03CWT. Each row sums to 100%.



The U-Net confusion matrix shows moderate thin-ice confusion with thick ice. The LSTM matrix shows high thick-ice accuracy but near-complete collapse on thin ice and open water when the track is sparse. The deep fusion matrix shows the most balanced per-class performance, with diagonal dominance across all three classes.

6. Ablation Study

Four fusion variants were trained on T02CNA and T02CNC and evaluated on T03CWT to isolate the contribution of each design decision.

Variant	mIoU	Macro F1	Configuration
fusion_v2	0.8020	0.8882	Photon branch from random init (collapsed to dominant class)
fusion_v3	0.8949	0.9431	Larger recurrent branch (hidden=128), random init
fusion_v4	0.8982	0.9453	Pretrained photon branch, frozen during fusion
deep_fusion	0.9010	0.9468	Pretrained photon branch, fine-tuned at 0.1x LR (best)

Key finding: the smaller pretrained-and-fine-tuned branch (hidden=96) outperforms the larger randomly initialized branch (hidden=128). Transferred photon representations contribute more than raw model capacity for this fusion task.

7. Key Findings

1. Deep fusion achieves mIoU 0.9010, surpassing U-Net (+3.1 pp) and LSTM (+20.3 pp).
2. Largest gains on minority classes: thin-ice IoU +4.5 pp, open-water IoU +3.6 pp over the optical baseline.
3. The LSTM photon branch alone struggles with thin ice (IoU 0.543) due to sparse ICESat-2 ground-track coverage, but provides strong height discrimination when fused with image features.
4. Transfer learning (pretrain then fine-tune) contributes more than increasing recurrent capacity.
5. Focal loss with asymmetric class weights ($\alpha=[0.05, 0.45, 0.60]$) is essential to prevent prediction collapse on the dominant thick-ice class.
6. Geographic generalization is confirmed: the held-out tile T03CWT is spatially separated from both training tiles.

8. Conclusions and Acknowledgments

Conclusions

This project demonstrates that deep feature-level fusion of Sentinel-2 optical imagery and ICESat-2 ATL03 photon altimetry substantially improves per-pixel Antarctic sea ice classification compared to either modality alone. The Squeeze-and-Excitation fusion module effectively recalibrates the contribution of each modality for different spatial regions, enabling accurate prediction even for spectrally ambiguous thin-ice pixels.

The 21-run LSTM hyperparameter sweep confirms the importance of class-reweighted focal loss and appropriate recurrent capacity for the photon-only baseline. The ablation study confirms that transferring and fine-tuning the pretrained photon branch is the most impactful single design decision in the fusion model.

Future directions include incorporating additional Sentinel-2 spectral bands, exploring attention-based cross-modal fusion, and extending to other polar regions and seasons.

Repository Structure

<code>deep_fusion.ipynb</code>	Deep fusion model (primary result)
<code>lstm_sweep.ipynb</code>	21-configuration LSTM hyperparameter sweep
<code>requirements.txt</code>	Python dependencies
<code>confusion_matrices/</code>	Green and blue confusion matrix PNGs
<code>runs/deep_fusion/</code>	Deep fusion outputs (mIoU 0.9010)
<code>runs/bilstm/</code>	LSTM-only outputs (mIoU 0.6978)
<code>archive/</code>	Superseded experiments and earlier fusion variants

Acknowledgments

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Data sources: ICESat-2 ATL03 products -- NASA National Snow and Ice Data Center (NSIDC); Sentinel-2 imagery -- ESA Copernicus Programme.

A corresponding manuscript is in preparation.